

Section 13 Solution

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3 Exercises

3.1 Profit Maximization with Taxes (Problem 2, Midterm 2, Spring 2012)

We consider the market for widgets, which is characterized by the aggregate (inverse) demand function $p(X) = a - bX$, where X is the total quantity of widgets demanded in the market. The cost function of each company is $c(y) = cy^\alpha$, with $c > 0$.

1. Assume perfect competition (that is, the price p of the widget is given) and set up the profit maximization of each firm.

Answer:

Each firm maximizes

$$\max_y py - c(y) = py - cy^\alpha$$

2. Solve for the profit-maximizing level of production $y^*(p)$ (that is, the supply function) using the first-order condition. **Answer:**

The first-order condition for y^* is $p - \alpha cy^{\alpha-1} = 0$ leading to the solution $y^* = (p/\alpha c)^{1/(\alpha-1)}$ (so long as α is different from zero)

3. Check the second-order conditions. Under what values of the parameters are they satisfied? Interpret the economic significance of this parameter restriction.

Answer:

The second-order condition for y^* is $-\alpha(\alpha-1)cy^{\alpha-2} < 0$.

This condition is satisfied for $\alpha > 1$. This condition is the condition of decreasing returns to scale.

4. Now consider the conditions for the market equilibrium. For points 4-7, assume that the parameters are such that the second order conditions are satisfied. Assume that N firms produce and write the equation for the equilibrium price p^* that equates aggregate supply and demand. Do not attempt to solve explicitly for p^* .

Answer:

To solve for the equilibrium price, we must equate supply Y and demand X . First, remember that total production is $Y = Ny^* = N(p/\alpha c)^{1/(\alpha-1)}$. One way to do this is to substitute the solution for Y^* from the production decision into the demand function:

$$p - a + bN \left(\frac{p}{\alpha c} \right)^{1/(\alpha-1)} = 0$$

5. Now introduce taxation. Denote by p the price inclusive of tax that the consumer pays, and by $p - t$ the price net of tax that accrues to the producer. Rewrite the market equilibrium condition.

*These notes are a consolidation of those of Jianmeng Lyu, Sam Leone, Pablo Munoz, Yassine Sbair Sassi, Jon Schellenberg, Nick Li, Katalin Springel, Jonas Tungodden, Dan Acland, Justin Gallagher, Mariana Carrera, Matt Leister, Ivan Balbuzanov, Anne Karing, and David Wu, Sam Wang, and Matteo Saccarola.

Answer:

The market equilibrium with taxes is

$$p - a + bN \left(\frac{p-t}{\alpha c} \right)^{1/(\alpha-1)} = 0$$

6. Use the implicit function theorem to compute $\partial p^*/\partial t$. **Answer:**

We can compute

$$\begin{aligned} \frac{\partial p^*}{\partial t} &= - \frac{-\frac{bN}{\alpha-1} \left(\frac{1}{\alpha c} \right)^{1/(\alpha-1)} (p-t)^{-\frac{\alpha}{\alpha-1}}}{1 + \frac{bN}{\alpha-1} \left(\frac{1}{\alpha c} \right)^{1/(\alpha-1)} (p-t)^{-\frac{\alpha}{\alpha-1}}} = \\ &= \frac{\frac{bN}{\alpha-1} \left(\frac{1}{\alpha c} \right)^{1/(\alpha-1)} (p-t)^{-\frac{\alpha}{\alpha-1}}}{1 + \frac{bN}{\alpha-1} \left(\frac{1}{\alpha c} \right)^{1/(\alpha-1)} (p-t)^{-\frac{\alpha}{\alpha-1}}} \end{aligned}$$

7. Show that $0 < \partial p^*/\partial t < 1$. What does it mean economically? **Answer:**

Both numerator and denominator are positive (remember $\alpha > 1$), hence $\partial p^*/\partial t > 0$. Since the denominator equals the numerator plus a positive number (1), the ratio has to be smaller than 1, hence $\partial p^*/\partial t < 1$. This shows that the incidence of the tax is partially on the consumer and partially on the producer.

3.2 Consumer Surplus (Problem 2, Midterm 2, Spring 2009)

We evaluate here the change in consumer surplus associated with a change in the price of good 1 from p_1 to p'_1 for a consumer with Cobb-Douglas utility $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, with $0 < \alpha < 1$.

1. Explain the intuition of why the change in consumer surplus is defined as $\Delta CS = e(p_1, p_2, u) - e(p'_1, p_2, u)$, where e is the expenditure function. (6 points)
2. Define the expenditure function. (4 points)
3. Derive an expression for the expenditure function $e(p_1, p_2, u)$ for this Cobb-Douglas case given that the Hicksian demands in this case are

$$\begin{aligned} h_1^*(p_1, p_2, u) &= u \cdot \left(\frac{p_2}{p_1} \cdot \frac{\alpha}{1-\alpha} \right)^{1-\alpha} \\ h_2^*(p_1, p_2, u) &= u \cdot \left(\frac{p_1}{p_2} \cdot \frac{1-\alpha}{\alpha} \right)^\alpha \end{aligned}$$

(5 points)

4. Solve for the change in consumer surplus $\Delta CS = e(p_1, p_2, u) - e(p'_1, p_2, u)$. (If you get stuck here, just skip to the next point) (5 points)
5. If you also substituted for u the expression for the indirect utility $v(p_1, p_2, M)$ you would get that ΔCS is proportional to

$$\left[1 - \left(\frac{p'_1}{p_1} \right)^\alpha \right] \cdot M \quad (1)$$

(Do not attempt to do this, it involves a fair amount of algebra, take it as given) Using expression (1), show that the following statements are true, and provide intuitive explanations for each of them: (i) $\Delta CS > 0$ if and only if $p'_1 < p_1$; (ii) holding constant p'_1 and p_1 with $p'_1 < p_1$, ΔCS is increasing in α ; (iii) holding constant p'_1 and p_1 with $p'_1 < p_1$, ΔCS is increasing in M . (15 points)

Answer:

1. $e(p'_1, p_2, u)$ is the minimum amount of money required to attain the initial utility level u at the new price of good 1 (p'_1). Subtracting this from the expenditure that is required before the price change ($e(p_1, p_2, u)$) gives the maximum amount the consumer would be willing to pay to enact the price change. If the consumer were to pay this amount (payment may be positive or negative), then she would reach an identical utility level after the price change. In the absence of this compensating payment, she will achieve either greater utility (if $p'_1 < p_1$) or less utility (if $p'_1 > p_1$), but we can monetize the value of this gained/lost utility as the amount she would pay to get it.

2. The expenditure function is defined as:

$$e(p_1, p_2, u) = \min_{h_1, h_2} p_1 h_1 + p_2 h_2 \text{ s.t. } u(x_1, x_2) \geq u$$

OR:

$$e(p_1, p_2, u) = p_1 h_1^* + p_2 h_2^*$$

where h_1^* and h_2^* are the quantities of goods 1 and 2 that minimize spending while attaining utility level u .

- 3.

$$\begin{aligned} e(p_1, p_2, u) &= p_1 h_1^* + p_2 h_2^* \\ &= p_1 * u \cdot \left(\frac{p_2}{p_1} \cdot \frac{\alpha}{1-\alpha} \right)^{1-\alpha} + p_2 * u \cdot \left(\frac{p_1}{p_2} \cdot \frac{1-\alpha}{\alpha} \right)^{\alpha} \\ &= u p_1^{\alpha} p_2^{1-\alpha} \left[\left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^{\alpha} \right] \\ &= u p_1^{\alpha} p_2^{1-\alpha} \left[\frac{\alpha^{-\alpha}}{(1-\alpha)^{1-\alpha}} \right] \end{aligned}$$

4. Using the expression found above,

$$\begin{aligned} \Delta CS &= e(p_1, p_2, u) - e(p'_1, p_2, u) \\ &= u p_2^{1-\alpha} \left[\frac{\alpha^{-\alpha}}{(1-\alpha)^{1-\alpha}} \right] (p_1^{\alpha} - p_1'^{\alpha}) \end{aligned}$$

By the fundamental theorem of calculus, the same expression can be found by integrating $\frac{\partial e(p_1, p_2, u)}{\partial p_1} = h_1^*$ over the range $p_1'^{\alpha}$ to p_1^{α} .

5. If we take the previous expression and substitute

$$u = v(p_1, p_2, M) = \left(\frac{\alpha}{p_1} \right)^{\alpha} \cdot \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} \cdot M,$$

we obtain

$$\begin{aligned} \Delta CS &= \left(\frac{\alpha}{p_1} \right)^{\alpha} \left(\frac{1-\alpha}{p_2} \right)^{1-\alpha} M p_2^{1-\alpha} \left[\frac{\alpha^{-\alpha}}{(1-\alpha)^{1-\alpha}} \right] (p_1^{\alpha} - p_1'^{\alpha}) = \\ &= \frac{1}{p_1^{\alpha}} M (p_1^{\alpha} - p_1'^{\alpha}) = M \left[1 - \left(\frac{p'_1}{p_1} \right)^{\alpha} \right] \end{aligned}$$

(You were not required to show this) (i) Proof of $\Delta CS > 0 \Leftrightarrow p'_1 < p_1$:

Given that ΔCS is proportional to the expression in (1), its sign is the sign of $\left[1 - \left(\frac{p'_1}{p_1} \right)^{\alpha} \right]$. This quantity is positive if and only if $\left(\frac{p'_1}{p_1} \right)^{\alpha} < 1 \Leftrightarrow \frac{p'_1}{p_1} < 1^{1/\alpha} \Leftrightarrow p'_1 < p_1$.

Intuition: Given Cobb-Douglas utility, the consumer is buying a positive quantity of x_1 at the original price. After a price decrease, s/he will be able to purchase the same bundle for less money and use the leftover money to buy more goods: the increase in utility that follows corresponds to an increase in CS.

(ii) Proof that ΔCS is increasing in α :

$$\frac{d\Delta CS}{d\alpha} \propto \frac{d \left[1 - \left(\frac{p'_1}{p_1} \right)^\alpha \right] \cdot M}{d\alpha} = -M \frac{d \left(\frac{p'_1}{p_1} \right)^\alpha}{d\alpha} = -M \left(\frac{p'_1}{p_1} \right)^\alpha \ln \left(\frac{p'_1}{p_1} \right)$$

Given that $p'_1 < p_1$, $\ln \left(\frac{p'_1}{p_1} \right) < 0$. Therefore, the quantity above is positive and $\frac{d\Delta CS}{d\alpha} > 0$.

Intuition: α is the share of income spent on x_1 . When more x_1 is purchased, the consumer has more to gain from a decrease in p_1 .

(iii) Proof that ΔCS is increasing in M :

$$\frac{d\Delta CS}{dM} \propto \frac{d \left[1 - \left(\frac{p'_1}{p_1} \right)^\alpha \right] \cdot M}{dM} = \left[1 - \left(\frac{p'_1}{p_1} \right)^\alpha \right]$$

From (i), this quantity is positive when $p_1 > p'_1$.

Intuition: A consumer with higher M buys a larger quantity of x_1 at p_1 , and thus gets more “extra” income when p_1 falls to p'_1 .